Introduction to Data Analytics
MS - 4610

Session 15 – Association Rule Mining (ARM)

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Association Rule Mining

- Association rule mining (ARM) has emerged as a powerful and specialized tool to identify patterns in large datasets.
- Originally developed to identify purchasing patterns of customers in transactions database at retail outlets.
- It is often referred to as market basket analysis
- A general Associative Rule is an implicative rule of the form $A \rightarrow B$

TID	*Items*
1	Bread, Milk,Coffee
2	Bread, Diaper, Baby Food, Eggs
3	Milk, Diaper, Baby Food, Coffee
4	Bread, Milk, Diaper, Baby Food
5	Bread, Milk, Diaper, Coffee

Example of associative rules:

{Diaper} $\rightarrow$ {Baby Food}
{Bread, Butter} $\rightarrow$ {Jam}

Definitions:

- Item set
- Antecedent
- Consequent

Association Rule Mining

- The rows of a data set represent the instances, and the columns correspond to a common set of attributes associated with the instances.
- ARM seeks to identify patterns or regularities between the attributes, based on the values they take up with relation to each other.
- A typical example of a pre-processed binary table which feeds into ARM algorithm which represents the presence or absence of an item in dataset.
- This can go well beyond a market-basket scenario
- So how is this analysis different from Clustering?

Customers	Gender	High income	High Expense	Owns Insurance	Lives in X	Age<30
T1	1	1	1	0	0	0
T2	1	0	0	1	1	1
T3	0	1	1	1	1	0
T4	1	1	0	1	1	0
T5	1	1	1	1	0	0

Association Rule Mining: challenges

- Candidate Generation and Candidate Evaluation
- Candidate Generation:
  - This step would involve sustainably mining all the possible or a set of interesting rules from all the possible item-sets in the dataset.
  - A basic brute force approach would evaluate all possible rules. Is that practical?
    - How many item sets? 2^d
    - How many rules?

$$R = \sum_{k=1}^{d-1}\left[\binom{d}{k} \times \sum_{j=1}^{d-k}\binom{d-k}{j}\right]$$
$$= 3^d - 2^{d+1} + 1$$

  - Break the problem into two parts:
    - Antecedents and consequents are composed of itemsets
    - The concept of Support P(of an itemset)

ARM: Candidate generation

- The *a*PRIORI algorithm. With example:
- Transactions list:
  {1,2,3,4},{1,2,4},{1,2},{2,3,4},{2,3},{3,4},{2,4}
- Pick some threshold

- The biases from *a*PRIORI. The rare items problem

Candidate Evaluation

- The step would be involve the evaluation of the strength/interestingness of the rules mined in the candidate generation phase.
- Metrics:
  - Lift: $P(AB)/(P(A).P(B))$
  - Confidence: $P(A|B)$
  - Accuracy: $P(AB) + P(\overline{AB})$
  - Precision, Recall and F-Measure
  - And 100 others at least : Tew, C., et al. "Behavior-based clustering and analysis of interestingness measures for association rule mining." *Data Mining and Knowledge Discovery* 28.4 (2014): 1004-1045.

Properties

- Objective Measures
  - Objective measures observe some mathematical properties
    - The O matrix:

	B	B’	
A	f11	f10	f1+
A’	f01	f00	f0+
	f+1	f+0	N

    - Properties (note: M refers to the measure):
      - M = 0 if A and B are statistically independent
      - M Monotonically increases with P(A,B) when P(A) and P(B) are kept same
      - M Monotonically decreases with increase in P(A) when P(B) and P(A,B) are unchanged
      - Variable permutation
      - Row and column scaling
      - Row or column permutation and Inversion Invariance
      - Null Invariance

    - Tan, Pang-Ning, Vipin Kumar, and Jaideep Srivastava. "Selecting the right objective measure for association analysis." *Information Systems* 29.4 (2004): 293-313.

Examples

- Expanding Lift: $P(AB)/(P(A).P(B))$

- How would you measure "*M Monotonically increases with P(A,B) when P(A) and P(B) are kept same*"
  - O(M2)>O(M1) If M2 = M1+[k –k; -k k]

- Showing the cosine is null invariant $P(AB)/(\sqrt{P(A).P(B)})$